

LE-HUYNH TRUC-LY

I have over five years of experience in data analysis across projects spanning environmental science, health science, political science, and biology. My passion lies in data analysis, visualization, automation, and creating digital tools for research. This very CV was even crafted using code.

PROJECTS

2024
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Present

● **Climate and Health in Virginia: Investigating Underlying Causes of Climate-Related Diseases and Disparities** 
Environmental Institute, University of Virginia  Virginia, USA

Skills used — Geospatial health modeling, Big data analytics, R, RMarkdown, Git, GNU Make, R package development
- Analyzing large-scale patient-level health records, environmental, and demographic datasets to identify unequal health risks during climate extremes, particularly extreme heat events

2025
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Present

● **Does Urban Green Space Mitigate Emergency Department Cases During Heat Waves?** 
Environmental Institute, University of Virginia  Virginia, USA

Skills used — Geospatial health modeling, Big data analytics, R, RMarkdown, Git, GNU Make, Parallel computing
- Analyzing large-scale patient-level health records, ERA5-Land reanalysis, and landscape metrics to quantify protective effects of neighborhood green spaces during extreme heat events

2024
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2025

● **Local-global linkages in biodiversity governance** 
Collaboration with Wyss Academy for Nature, University of Bern  Bern, Switzerland

Skills used — Network analysis, Text Mining analysis, Discourse analysis, R, RMarkdown, Git, GNU Make
- Analyzing large-scale governance network data to assess whether the global biodiversity regime delivers transformative change

2023
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Present

● **Toward personalized acute myeloid leukemia (AML)** 
Collaboration with Biotech Research & Innovation Centre, University of Copenhagen  Copenhagen, Denmark

Skills used — Bioinformatics, R, R package development, RMarkdown, Git, GNU Make
- Analysed mass spectrometry-based proteomics data from a genetically engineered AML mouse model to identify novel druggable oncogenic signaling targets
- Analysed patient proteomics data to identify clinically actionable signaling targets for personalized AML treatment

CONTACT

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SKILLS

Data Science: Biostatistics, Geospatial health modeling, Bayesian modeling, Big data analytics

Programming: R (advanced: published 4 R-packages , Python, SAS, JAGS, Stan

Reproducible Report: Markdown/RMarkdown, Quarto, LaTeX

DevOps: Git, GNU Make, Parallel computing

Visualization: ggplot2, plotly, leaflet, Shiny

Laboratory: Molecular Biology, Molecules Analysis, Water Quality Analysis

Languages: Vietnamese (native), English (fluent), Japanese (elementary)

- 2021
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2022
- **Development of predictive biomarker spectrum for neuropsychiatric systemic lupus erythematosus (NPSLE)** [🔗](#)
Collaboration with Graduate School of Biomedical Sciences, Nagasaki University 📍 Nagasaki, Japan

Skills used — Bayesian modeling, Biostatistics, R, SAS, RMarkdown, LaTeX, Git, GNU Make
- Estimated the cutoff concentration of anti-suprabasin antibodies in a huge patient database to predict NPSLE, through utilizing a novel Bayesian model
- Implemented reproducible workflow for data analysis and data visualization
- 2020
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2023
- **Bayesian predictive model for toxic cyanobacteria occurrence from eutrophication and climate data** [🔗](#)
Graduate School of Engineering, Nagasaki University 📍 Nagasaki, Japan

Skills used — Bayesian modeling, R, JAGS, Stan, RMarkdown, LaTeX, Git, GNU Make, Parallel computing
- Developed Bayesian model addressed zero-inflation issue in cyanobacterial data, predicting presence probability, abundance, and WHO alert level exceedance for toxic cyanobacteria
- Improved overall modelling efficiency through the implementation of Parallel Processing techniques
- Automated the data analysis and reporting system using GNU Make, R, and Rmarkdown

EDUCATION

- 2018
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2023
- **Doctor of Philosophy - PhD in Engineering (Water and Environmental Science)**
Nagasaki University 📍 Nagasaki, Japan

• Dissertation: Statistical Investigation into the Effects of Climate and Eutrophication on the Occurrence of Cyanobacteria in Small Ponds and Reservoirs
- 2016
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2018
- **Master of Engineering - MEng in Water and Environmental Engineering**
Nagasaki University 📍 Nagasaki, Japan

• Thesis: Statistical Analysis on the Relationship among Environmental Factors, Microcystin Synthesis Gene, and Microcystin Degradation Gene
- 2012
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2016
- **Bachelor of Science - BS in Environmental Engineering Technology**
Vietnam National University – Ho Chi Minh City (VNU-HCM) University of Science 📍 Ho Chi Minh City, Vietnam

• Thesis: Research on Sewage Sludge Dewatering System using Solar Energy

WORK EXPERIENCE

- **Postdoctoral Research Associate**
Environmental Institute, University of Virginia, USA 📍 2024 - Present
- **Visiting Researcher**
Graduate School of Engineering, Nagasaki University, Japan 📍 2024 - 2025
- **Independent Data Analyst**
Self-employed 📍 2021 - Present

• Services: Reproducible data analysis (in environmental science, health science, biology, bioinformatics), data visualization, cartography, data communication, consulting, R package development
• Tools: R, SAS, Python, ggplot2, plotly, leaflet, Shiny

- **Project Researcher**
Graduate School of Engineering, Nagasaki University, Japan 📍 2023
- **Research Assistant**
Biological Treatment and Ecological Engineering Laboratory, Nagasaki University, Japan 📍 2018 - 2019
- **Technical Assistant**
Biological Treatment and Ecological Engineering Laboratory, Nagasaki University, Japan 📍 2017 - 2019
- **Graduate Teaching Assistant**
Graduate School of Engineering, Nagasaki University, Japan 📍 2018 - 2020

ADDITIONAL INFORMATION

● R Package Development

- **chva.extras**: Supplementary tools for climate and health research in Virginia. **Le-Huynh, T.-L.**, Davis, R. E., Novicoff, W. M., DeGuzman, P. B. (2025) [🔗](#)
- **omics4drug**: An R toolkit to facilitate Mass Spectrometry-based Proteomics and Phosphoproteomics data analysis. Nguyen, T.-K. Y., **Le-Huynh, T.-L.** (2025) [🔗](#)
- **cellviability**: An R toolkit for analysis of cell-based drug screening experiments. Nguyen, Y. T.-K., **Le-Huynh, T.-L.**, Theilgaard-Monch, K. (2026) [🔗](#)
- **lehuynh**: Personal R package. **Le-Huynh, T.-L.** (2023) [🔗](#)

● Honours & Awards

- Postdoctoral Climate Fellowship, University of Virginia, USA (2024 – present)
- Planetary Health Research Fellowship, Nagasaki University, Japan (2022 – 2023)
- Asian Student Foundation Scholarship, Asian Student Foundation, Japan (2017 – 2019)
- Monbukagakusho Honors Scholarship, Japan Student Services Organization, Japan (2016 – 2017)
- CHEER for Vietnam Scholarship Award for Innovation and Creativity, CHEER for Vietnam Organization, USA (2015)
- Academic Excellence Award, Vietnam National University – Ho Chi Minh City University of Science, Vietnam (2014)

● Funded Grants

- **Spark Grant, Investigator**, funded by Environmental Institute, University of Virginia for project *Does Urban Green Space Mitigate Emergency Department Cases During Heat Waves?* [🔗](#)

● Certifications & Training

- Bayesian Modeling for Environmental Health, Columbia University Mailman School of Public Health, USA (2026)
- Community-Engaged Environmental Data Science, Columbia University Mailman School of Public Health, USA (2026)

● Service & Community Engagement

- Peer reviewer for Journal Of Exposure Science And Environmental Epidemiology (2026), Respiratory Research (2026)
- Translated R cheat sheets (Git & GitHub, gtsummary, RStudio IDE) into Vietnamese [🔗](#)
- Member of European Geosciences Union - EGU (2026)
- Member of National Center for Faculty Diversity and Development, USA (2025 - present)
- Member of Virginia Women in High Performance Computing, USA (2024 - present)
- Member of US National Postdoctoral Association, USA (2024 - present)
- Member of Japan Society on Water Environment, Japan (2017 - 2023)

● Selected Publications

- **Le-Huynh, T.-L.**, Novicoff, W. M., DeGuzman, P. B., Davis, R. E. (2026). Assessing the Importance of Spatial Scale in Climate-Health Models: the Effect of Temperature on Emergency Department Visits in Richmond, Virginia. *GeoHealth*, 10, e2026GH001852.
- Nguyen, V. T. H., **Le-Huynh, T.-L.**, Gadola, S., Nguyen, Q., Walters, G., Owuor, M. A (2025). Local-Global Linkages in Biodiversity Governance: The Regime Complex of the Convention on Biological Diversity Agenda for Nature Pledges. *Environmental Science & Policy*, 173(November), 104246.
- Saenchan, S., Shimizu, K., Iwami, N., Angalika, M. W. S., Maseda, H., Minh, N. B., Vu, H. V., **Le-Huynh, T.-L.**, Itayama, T. (2025). Effects of pH and ultrasonication for decomposition of Microcystis formed colony in a bloom sample from a reservoir. *Desalination and Water Treatment*, 322, 101203.
- Huynh, V. V., Nguyen, M. B., Somsri, S., **Le-Huynh, T.-L.**, Ueyama, T., Shirayanagi, S., Itayama, T. (2023). Investigation of performance in MBR operated with low DO for low C/N ratio wastewater. *Water, Air, & Soil Pollution*. 235(540).
- **Le-Huynh, T.-L.**, Itayama, T., Mitsunaga, K., Angalika, M., Suzuki, S. (2022). Application of hurdle Poisson model to predict the abundance of toxic cyanobacteria Microcystis in reservoirs. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 47–51.
- **Le-Huynh, T.-L.**, Iwami, N., Whangchai, N., Gutierrez, R., Shimizu, K., Itayama, T. (2022). Statistical analysis of the effects of environmental factors and fish species on class-sorted phytoplankton composition in aquaculture ponds in northern Thailand. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 32–38.
- Angalika, M. W. S., Suzuki, S., **Le-Huynh, T.-L.**, Itayama, T., Tanaka, W. (2022). Assessing nutrient budget of ungauged catchment using intermittent water quality markers. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 1–10.
- Hoang, T. T. T., Ichinose, K., Morimoto, S., Furukawa, K., **Le-Huynh, T.-L.**, Kawakami, A. (2022). Measurement of anti-suprabasin antibodies, multiple cytokines and chemokines as potential predictive biomarkers for neuropsychiatric systemic lupus erythematosus. *Clinical Immunology*, 237(March), 1–8.

● Selected Scientific & Professional Presentations

- **Le-Huynh, T.-L.**, Novicoff, W., DeGuzman, P., Davis, R., Heat and Emergency Department Visits: Does Social Vulnerability Matter? [Oral presentation], *2026 Climate Fellows Research Showcase*, Charlottesville, Virginia, USA, September 2026.
- **Le-Huynh, T.-L.**, Davis, R. E., Novicoff, W. M., Enfield, K. B., Are Fireworks Associated with Respiratory Emergency Department Visits in Virginia? [Oral presentation], *24th International Congress of Biometeorology*, Novi Sad, Serbia, July 2026.
- Kakatkar, S., **Le-Huynh, T.-L.**, Davis, R. E., Heat, Air Quality, and Emergency Department Diagnoses: Social Vulnerability and Biometeorological Stress in Virginia [Oral presentation], *24th International Congress of Biometeorology*, Novi Sad, Serbia, July 2026.
- Davis, R. E., **Le-Huynh, T.-L.**, Novicoff, W. M., DeGuzman, P., Assessing Spatial Scale in Climate-Health Research [Oral presentation], *24th International Congress of Biometeorology*, Novi Sad, Serbia, July 2026.
- Ha, J., Hosseini, P., Davis, R., **Le-Huynh, T.-L.**, Novicoff, W., Delaar, G., Yoo, J., & Sa. H., Urban green space and emergency department visits during heat waves [Oral presentation], *Council of Educators in Landscape Architecture*, Portland, Oregon, USA, March 2026.
- **Le-Huynh, T.-L.**, Novicoff, W., DeGuzman, P., Davis, R., Exploring climate-health disparities in Virginia: Toward AI-driven solutions [Poster presentation], *Environmental Futures Forum 2025*, Charlottesville, Virginia, USA, October 2025.
- **Le-Huynh, T.-L.**, Iwami, N., Praphrute, R., Whangchai, N., Gutierrez, R., Shimizu, K., Itayama, T., Statistical analysis on phytoplankton population at hypertrophic ponds in northern Thailand [Oral presentation], *The 57th Annual Conference of Japan Society on Water Environment*, Ehime, Japan, March 2023.
- **Le-Huynh, T.-L.**, Mitsunaga, K., Itayama, T., A Bayesian model for predicting the growth of toxic Microcystis from air temperature and trophic state index [Oral presentation], *The 3rd International Conference on Renewable Energy, Sustainable Environmental and Agricultural Technologies*, Chiangmai, Thailand, December 2021.
- **Le-Huynh, T.-L.**, Itayama, T., Nguyen, T. H. G., Xia, D., Shimizu, K., Iwami, N., Okano, K., Maseda, H., Praphrute, R., Ruangdet, K., Gutierrez, R., Whangchai, N., Influence of environmental factors on Microcystins degradation bacteria and toxigenic cyanobacteria bloom: a Bayesian approach [Poster presentation], *The NaToxAq Conference on Natural Toxins: Environmental Fate & Safe Water Supply*, Brno, Czech Republic, September 2020.